

Social Choice Theory

ocial Choice Theory is a branch of economics, political science, and game theory that studies **how individual preferences can be combined to make a collective decision**.

In simple terms:

Social Choice Theory asks: "If many people have different opinions, what is the fairest way to make one decision for everyone?"

It provides mathematical methods to aggregate individual choices into a single social outcome

Formal Definition

Social Choice Theory is the study of **collective decision-making**, where the preferences of individuals are combined using a voting or decision rule to produce a social preference or collective choice.

It is the theoretical foundation of:

- Voting systems
- Elections
- Committee decisions
- Public policy
- Democratic institutions
- Resource allocation
- AI decision-making
- Multi-agent systems

Famous Theorems

Arrow's Impossibility Theorem

No perfect voting system exists.

Gibbard–Satterthwaite Theorem

Every reasonable voting system with at least three choices can be strategically manipulated under certain conditions.

May's Theorem

Majority voting is the fairest method for choosing between **two alternatives**, as it satisfies key fairness properties such as equal treatment of voters and neutrality between options.

Advantages

- Encourages democratic decision-making.
 - Provides mathematical methods for fair aggregation.
 - Helps design better voting systems.
 - Improves transparency in collective decisions.
 - Widely applicable in economics, politics, and computer science.
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Limitations

- No voting rule is perfectly fair (Arrow's theorem).
- Voting systems can be manipulated strategically.
- Majority decisions may ignore minority interests.
- Preference cycles can prevent a clear winner.
- Different voting methods may produce different outcomes from the same preferences.